Helix Thready — Prototypes & Motion (Figma · PenPot · Lottie)

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Rev	Date	Author	Change
1	2026-07-21	swarm (design)	Initial complete draft: Figma plan, interactive vs. non-interactive prototypes, motion/transition spec, export pipeline (Figma/ PenPot/PDF/PNG/SVG/Lottie), review loop, Docs Chain wiring
2	2026-07-22	swarm (design · Pass 3)	Depth pass to parity with the rest of the area: **corrected the unverified "first-party Figma plugin" claim** to an explicit assumption + a source-confirmed fallback bridge (§2, §7); added the **interactive-prototype coverage & traceability matrix** (§4.1, every journey → screens → flow → states/ motion → surfaces); added **prototype verification & runtime evidence** (§8.1, the anti-bluff gate on the review loop); enumerated the design-relevant subset of the 15 mandated test types the prototypes exercise
3	2026-07-22	swarm (design · package critic)	Wired the plan to the **materialized artifacts now on disk** (new §1.1): the interactive prototype (`screens/web/index.html` + the 30 rendered screens), the non-interactive set (`exports/png/` 49 renders light+dark 2×, `exports/design-book.pdf` 54 pp), the Lottie catalogue + self-contained `motion/preview.html`, and the PenPot/Figma-IR hand-off bundles. Plan sign-off verified in DESIGN_PACKAGE_REPORT.md.

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1. Requirements (verbatim intent)

From the request (§Design) `[OPERATOR]`:

- **Full Figma design + interactive AND non-interactive prototypes** for every visual client.
- **Designed with OpenDesign** `[CONSTITUTION §11.4.162]`, refined by **multiple independent-agent reviews** (like code review) until UI/UX perfection.
- Materials in **all major formats** — **PenPot, PDF, PNGs, and lots of Lottie animations**.
- **Stunning transition effects** between screens/pages and during UI interactions.
- **Full forms validation, hints, tooltips**; a unique design created exclusively for Thready.
- Materials **connected to the project via all hooks and Docs Chain**, so a change that affects client UI/UX propagates.
- **Versioned** like all documentation.

1.1 Materialized artifacts on disk

This plan is no longer prospective — the deliverables it specifies are **rendered and present in the area** (audited in DESIGN_PACKAGE_REPORT.md). Map from mandate to concrete artifact:

Mandate	Materialized artifact (VERIFIED on disk)
Interactive prototype	`screens/web/index.html` — a journey-walking shell that embeds the live rendered screens (all 14 web sibling links resolve); the 30 screens across web/mobile/desktop/TUI/marketing are self-contained, light + dark
Non-interactive prototype	`exports/png/` — 49 PNG renders (light + dark, 2× deviceScaleFactor) · `exports/design-book.pdf` — 54-page portable design book
Lottie motion + transitions	`motion/` — 6 Lottie `.json` (incl. `transition-fade-slide.json`), + the self-contained `motion/preview.html` board (vendored lottie-web, zero network)
PenPot / Figma hand-off	`exports/penpot/` (16 SVG + 22 PNG) · `exports/figma/` — **28** genuine `.od-figma.json` IR captures (11 291 nodes) + `IMPORT.md`

The **cloud** side of Figma/PenPot hand-off (a live `.fig`, a hosted PenPot project) is **operator-only** and honestly deferred — see §7 and the report's operator-action list; nothing here claims a cloud push occurred.

2. Tooling: OpenDesign + Figma (+ PenPot, Lottie)

`[CONSTITUTION + request]` `[Q26]`:

- **OpenDesign** (`nexu-io/open-design` , release 0.13.0 `[VERIFIED]`) is the design-system **source of truth** that drives tokens and generates artifacts. A **first-party Figma plugin** for the token→Figma sync is **assumed but NOT source-confirmed** — only `export.ts` emitting `tokens.json` / `theme.json` /CSS/ PPTX/PDF was verified `[VERIFIED — export.ts]`. Treat the plugin as `[ASSUMPTION — verify at integration]`; if it does not exist, the **source-confirmed fallback** is to import `tokens.json` into **Figma Variables** (the tokens map 1:1 to Figma variable collections/modes, §3), so the design→code token sync does not depend on an unverified plugin.
- **Figma** is the **source of record** for the component set and screen frames + the interactive prototype layer (in-house standard `[IN-HOUSE]`).
- **PenPot** — required by the request but **not currently used anywhere in the org**; introduced for Thready. **PenPot is not a native OpenDesign export target** `[VERIFIED — export.ts emits tokens.json/theme.json/CSS/PPTX/PDF only]` → a bridge is required `[OPEN: THREADY-DES-02]`.
- **Lottie** — required (motion), also **not a native OpenDesign/Figma-plugin export**; produced via an After Effects/ `bodymovin` or a Figma-to-Lottie plugin path → tracked in §7.

> **Honesty.** OpenDesign natively emits `tokens.json` , `theme.json` (antd), CSS vars, and screenshot-backed **PPTX/PDF**. **PDF/PNG/SVG** are reachable (PDF native; PNG/SVG via Figma or `svgo` / `sharp`). **PenPot and Lottie need an added bridge** — this doc specifies the bridge; it does not pretend the export is one click today.

3. Figma source-of-record structure

`[DEFAULT — adjustable]` — one Figma project, four files:

Figma file	Contents
Thready · Foundations	Token styles (colors from `thready` theme, type, spacing, radius, elevation, motion), light+dark variables, grid
Thready · Components	The component set (§ component-library.md §5) — every composite as a Figma component with all variants/states, bound to the token styles
Thready · Screens	Every screen frame from wireframes.md at phone/tablet/desktop, light+dark, per surface (Web, Desktop, Mobile per platform)
Thready · Flows	The interactive prototypes wiring screens into the four journeys (ux-flows.md)

Figma **variables/modes** carry the two theming axes (mode: light/dark; brand: Thready + a white-label sample) so a reviewer flips a mode instead of duplicating frames — mirroring the runtime token model exactly.

4. Interactive prototypes

Clickable, transition-rich prototypes for each **key journey** (the four in ux-flows.md), per surface:

Prototype	Covers	Surfaces
Add channel	wizard steps, sign-in, resolve preview, live sync start	Web, Mobile
Process post	queue → per-step progress → completion/retry (live states mocked)	Web, Mobile, TUI (recorded)
Search	query → semantic/keyword/hybrid → results → open	Web, Mobile
Manage account	switch, invite, branding editor + AA meter, billing	Web

Each interactive prototype demonstrates: real state transitions (empty→skeleton→content), form validation/hints/tooltips, error/retry, theme toggle (light↔dark), language switch (en/ru/sr-Cyrl), and the motion spec (§6). They are the artifact the independent-agent review (§8) evaluates.

4.1 Interactive/prototype coverage & traceability matrix

Each prototype is not free-standing — it traces to the exact **wireframe screens** (structural contract) and the **UX flow** (behavioral contract) it makes clickable, and it must exercise the full interaction-state set (the shared legend, wireframes §1.1) and the motion tokens it invokes (§6). This matrix is the checklist the review gate (§8) scores against, so no journey ships with a missing state or an off-spec transition `[DEFAULT — adjustable]`:

Journey (prototype)	Wireframe screens	Flow (source of truth)	States exercised	Motion (§6)	Surfaces
Add channel	§3.4 wizard + §3.11 messenger sign-in	ux §2 + §2.1	default · validating (Resolve) · resolved-preview · error (`422` / `403`) · success (optimistic add)	disclosure/ step slide, toast enter	Web, Mobile
Process post	§3.6 post detail + Dashboard queue	ux §3 + §3.1 reprocess	queued · running (per-step %) · done · failed+retry · `409` already-running	processing pulse (looped Lottie), progress fill	Web, Mobile, TUI (recorded)
Search	§3.7 search	ux §4	idle (recent) · skeleton · content · empty · degraded (`--warn`, hash-embedder `[GAP: 2.1]`) · timeout	skeleton→content cross-fade	Web, Mobile
Manage account	§3.10 admin + §3.11 branding	ux §5	default · dirty · previewing · validating · `422` AA-fail · success (audit-logged)	drawer slide, live-preview re-tint, toast	Web

Two honesty constraints the matrix carries forward from the wireframes/flows: the **Process-post** and **Search** prototypes must render the *stub-degraded* states (MeTube poll-only `[GAP: 6.5]`, no-OCR `[GAP: 2.6]`, hash-embedder `[GAP: 2.1]`) — a prototype that only shows the happy path would misrepresent what ships today. The **TUI** track for Process-post is a *recorded* walkthrough (asciinema/VHS), not a Figma click-through, because the TUI's realization is Lipgloss, not Figma (wireframes §5).

5. Non-interactive prototypes

Static, annotated deliverables (for spec/handoff/docs):

- **Redlines / measure specs** — spacing, sizes, token references per screen.
- **State sheets** — every component's states side-by-side (default/hover/focus/active/disabled/ error) in light + dark.
- **Screen catalog** — PDF/PNG of every frame, per surface + breakpoint.
- **Annotated flows** — the ux-flows diagrams overlaid on real frames (the request's "Figma exported screens with additional layers over it" for documentation).

6. Motion & transition spec

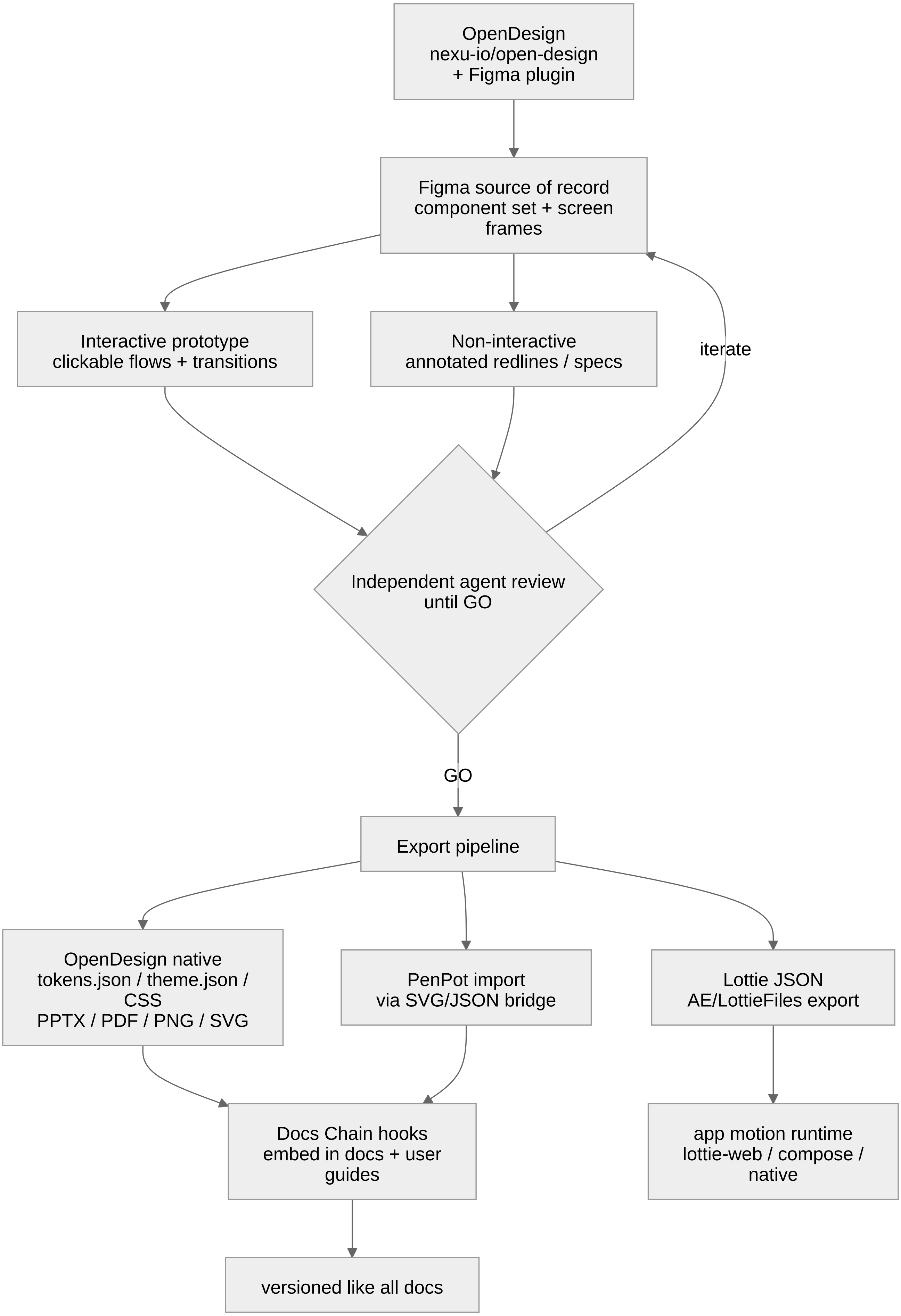
Grounded in the two shipped motion tokens (§ design-system.md §5) and extended for choreography `[DEFAULT — adjustable]`:

Interaction	Motion	Duration / easing
Button/control state	color/shadow	`--motion-fast` 150ms · `--ease-standard`
Disclosure / drawer / sheet	slide + fade	`--motion-base` 200ms · standard
Route/page transition	shared-axis / fade-through	250–300ms `[DEFAULT]`
Skeleton → content	cross-fade + subtle rise	200ms
Processing pulse	looped Lottie (indeterminate)	seamless loop
Toast enter/exit	slide-in + fade	200ms in / 150ms out
Theme toggle	crossfade tokens	150ms (instant if reduced-motion)

Lottie usage ("lots of Lottie animations"): the launcher-spiral loader (the Thready spiral drawing itself), the processing indeterminate pulse, empty-state illustrations, success checkmarks, and onboarding accents. Every animation ships a **static fallback** and is **skipped/frozen under `prefers-reduced-motion`** — motion is enhancement, never required for meaning.

```
// Lottie asset manifest (excerpt) [DEFAULT – adjustable]
{
  "animations": [
    { "id": "spiral-loader",     "loop": true,  "reducedMotionFallback": "spiral-static.svg" },
    { "id": "processing-pulse",  "loop": true,  "reducedMotionFallback": "pulse-static.svg" },
    { "id": "empty-channels",    "loop": false, "reducedMotionFallback": "empty-channels.svg" },
    { "id": "success-check",     "loop": false, "reducedMotionFallback": "check.svg" }
  ],
  "runtime": { "web": "lottie-web", "compose": "lottie-compose", "ios": "lottie-ios" }
}
```

7. Export pipeline


OpenDesign
nexu-io/open-design
+ Figma plugin
Figma source of record
component set + screen
frames
Interactive prototype
clickable flows + transitions
Non-interactive
annotated redlines / specs
iterate
Independent agent review
until GO
GO
Export pipeline
OpenDesign native
tokens.json / theme.json /
CSS
PPTX / PDF / PNG / SVG
PenPot import
via SVG/JSON bridge
Lottie JSON
AE/LottieFiles export
Docs Chain hooks
embed in docs + user
guides
app motion runtime
lottie-web / compose /
native
versioned like all docs

> Rendered PNG/SVG exported via Docs Chain (§11.4.65). Source: `diagrams/prototype-export-pipeline.mmd`.

Explanation (for readers/models that cannot see the diagram). OpenDesign seeds tokens into Figma — via its Figma plugin **if that plugin exists** (assumed, not source-confirmed, §2), otherwise via a `tokens.json` → Figma Variables import — making Figma the source of record for the component set and screen frames. From Figma two prototype tracks branch: the interactive (clickable flows with transitions) and the non-interactive (annotated redlines/specs). Both feed the independent-agent review gate, which iterates back into Figma until it returns GO (§8). On GO, the export pipeline runs. Native OpenDesign/Figma exports produce `tokens.json`, `theme.json`, CSS vars, and screenshot-backed PPTX/PDF, plus PNG/SVG. Two required formats are **not native** and go through added bridges: **PenPot** (imported via an SVG/JSON bridge) and **Lottie** (JSON exported via an After Effects/`bodymovin` or Figma-to-Lottie path). The native + PenPot artifacts flow into Docs Chain hooks so they embed into the docs and user guides and are **versioned like all documentation**; the Lottie JSON flows into the app motion runtimes (`lottie-web` on web, `lottie-compose` / `lottie-ios` on mobile). The dotted truth here is the two bridges — they are the tracked open item `THREADY-DES-02`.

Export target matrix `[DEFAULT — adjustable]`:

Format	Source	Native?	Use
`tokens.json` / `theme.json`	OpenDesign `export.ts`	✅	design→code token sync
CSS custom properties	OpenDesign `tokensToCssVars`	✅	the running theme
PDF	OpenDesign / Figma	✅	screen catalog, spec handoff, docs
PNG	Figma / `sharp`	✅	docs illustrations, store assets
SVG	Figma / `svgo`	❌	icon, scalable illustration
PPTX	OpenDesign (screenshot-backed)		stakeholder decks
PenPot	SVG/JSON **bridge**	bridge ❌	open-source design portability `[OPEN: THREADY-DES-02]`
Lottie JSON	AE `bodymovin` / Figma-to-Lottie	added	in-app motion `[OPEN: THREADY-DES-02]`

8. Review loop (design ≈ code review)

The request mandates design review “like we are doing with code review, until absolute perfection”. Mirror the code-review process `[CONSTITUTION §11.4.209]`:

- Each prototype/screen set is reviewed by an **independent agent** against an explicit rubric: a11y (WCAG AA, keyboard, SR), token fidelity (no off-token values), state completeness (empty/skeleton/error/validation), responsive integrity (phone/tablet/desktop), light+dark parity, motion-with-meaning + reduced-motion, localization (en/ru/sr-Cyrl incl. Cyrillic width), and brand correctness (no letters in the icon, attribution present).
- Findings iterate back into Figma; the gate returns **GO** only when the rubric passes — the same “iterate to GO” `[CONSTITUTION §11.4.134]` used for code.
- Visual-regression (`ScreenDiff` + `VisualRegression`) locks the approved frames so later drift is caught `[GAP: 9.3]`.

```yaml
# design-review rubric gate (excerpt)
review:
  target: interactive-prototype/add-channel
  checks:
    a11y_AA: pass            # contrast, focus, keyboard, SR names
    token_fidelity: pass      # every color/space value is a token
    states: [default, empty, skeleton, error, validation]  # all present
    responsive: [phone, tablet, desktop]
    themes: [light, dark]
    reduced_motion: honored
    i18n: [en, ru, sr-Cyrl]
    brand: icon-has-no-letters, helix-attribution-present
  decision: GO | ITERATE
```

8.1 Prototype verification & runtime evidence (anti‑bluff)

The review gate above is a **rubric**, but per the quality bar `[CONVENTIONS §7]` and the HelixQA anti-bluff posture `[CONSTITUTION §11.4.27]` a `GO` is only credible with **runtime evidence** — a green rubric row must be backed by an artifact, not an assertion. The prototype phase therefore produces, per journey × surface × theme:

- **Screenshots** of every state in the §4.1 matrix (default/skeleton/empty/error/validating/success), light **and** dark, captured from the clickable prototype — the same cells the implementation's `ScreenDiff` bank (design-system §8) will later lock, so the prototype and the built UI are diffed against **one** baseline.
- **Recorded interaction walkthroughs** (the interactive prototype click-path; asciinema/VHS for the TUI track) proving the transitions actually fire and honor `prefers-reduced-motion`.
- **A contrast/a11y report** per frame (the AA ratios the branding editor and the token themes claim, re-asserted on the rendered pixels, not just the token file).

This makes the prototypes a **testable predecessor** of the component library, not throwaway art: the frames feed the visual-regression baseline (`THREADY-DES-VR-01`, `[GAP: 9.3]`) and the adversarial **Challenges** decks (`THREADY-DES-CHAL-01`, a mandated test type — a 40-step reply chain, an all-`failed` pipeline, RTL/Cyrillic overflow, a 4.5:1-boundary accent), so a prototype that "looks done" is only accepted when its evidence exists. The design-relevant subset of the **15 mandated test types** `[CONSTITUTION §11.4.27]` the prototype phase exercises: **visual-regression**, **accessibility**, **interaction/e2e** (the clickable flows), **localization** (en/ru/sr-Cyrl width), **responsive** (phone/tablet/desktop), and **Challenges** scenario banks; the remaining mandated types (unit, integration, security, performance/load, etc.) attach to the *implementation* the prototype specifies, in component-library §9 and ../testing/index.md.

9. Docs Chain wiring & versioning

Per the request ("connected … through Docs Chain … so any change … gets applied") and `[CONSTITUTION §11.4.65/106]`:

- Exported PNG/SVG/PDF frames are committed under `design/exports/` and **referenced** by the docs and user guides; a Docs Chain context re-propagates on change.
- The `.mmd` diagram sources in `design/diagrams/` render to PNG/SVG via Docs Chain (already the convention for every diagram in this area).
- Design artifacts are **versioned like all docs** (revision headers, all-upstreams push §2.1); Figma file versions are tagged to the doc revision so a doc rev pins the design it describes.

```yaml
# .docs_chain/contexts/design.yaml (proposed)  [DEFAULT — adjustable]  [GAP: 10.1 docs_chain tooling]
context: design
watch:
  - docs/public/research/mvp/design/**/*.md
  - docs/public/research/mvp/design/diagrams/**/*.mmd
  - docs/public/research/mvp/design/exports/**
emit: [html, pdf]           # md ↔ HTML/PDF siblings (needs pandoc/weasyprint on host)
on_change: repropagate      # a token/frame change updates dependent docs
```

> **Honesty.** Docs Chain honestly **SKIPs** md→HTML/PDF when `pandoc` / `weasyprint` are absent (as on the current authoring host) `[GAP: 10.1 docs_chain]`; provision them on the dev host to emit siblings.

10. Deliverables checklist

A prototype phase is **done** only when all are present and GO-reviewed:

- [ ] Figma: Foundations + Components (all states) + Screens (all surfaces × breakpoints × light/dark) + Flows.
- [ ] Interactive prototypes for the four journeys × surfaces.
- [ ] Non-interactive: redlines, state sheets, screen catalog, annotated flows.
- [ ] Motion: Lottie set (§6) with static + reduced-motion fallbacks.
- [ ] Exports: PDF, PNG, SVG (native); **PenPot** + **Lottie** via bridge.
- [ ] Independent-agent review = GO; visual-regression frames locked.

- [ ] Docs Chain wired; artifacts versioned; slogan + attribution present everywhere.

11. Gaps & open items

- `[OPEN: THREADY-DES-02]` — **PenPot + Lottie bridges** are BUILD-NEW (not native OpenDesign exports). Workable item THREADY-DES-EXPORT-01: build/validate both bridges.
- `[GAP: 9.3 VisualRegression family]` — CI needed to lock approved frames (THREADY-DES-VR-01).
- `[GAP: 10.1 docs_chain]` — provision pandoc/weasyprint so exports+docs get HTML/PDF siblings.
- `[OPEN: THREADY-DES-04]` — Cyrillic width/subset must be validated in the review rubric.
- `[OPEN: THREADY-DES-13]` — decide PenPot's role: mirror of Figma (portability only) vs. a co-equal editing surface (higher maintenance).

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